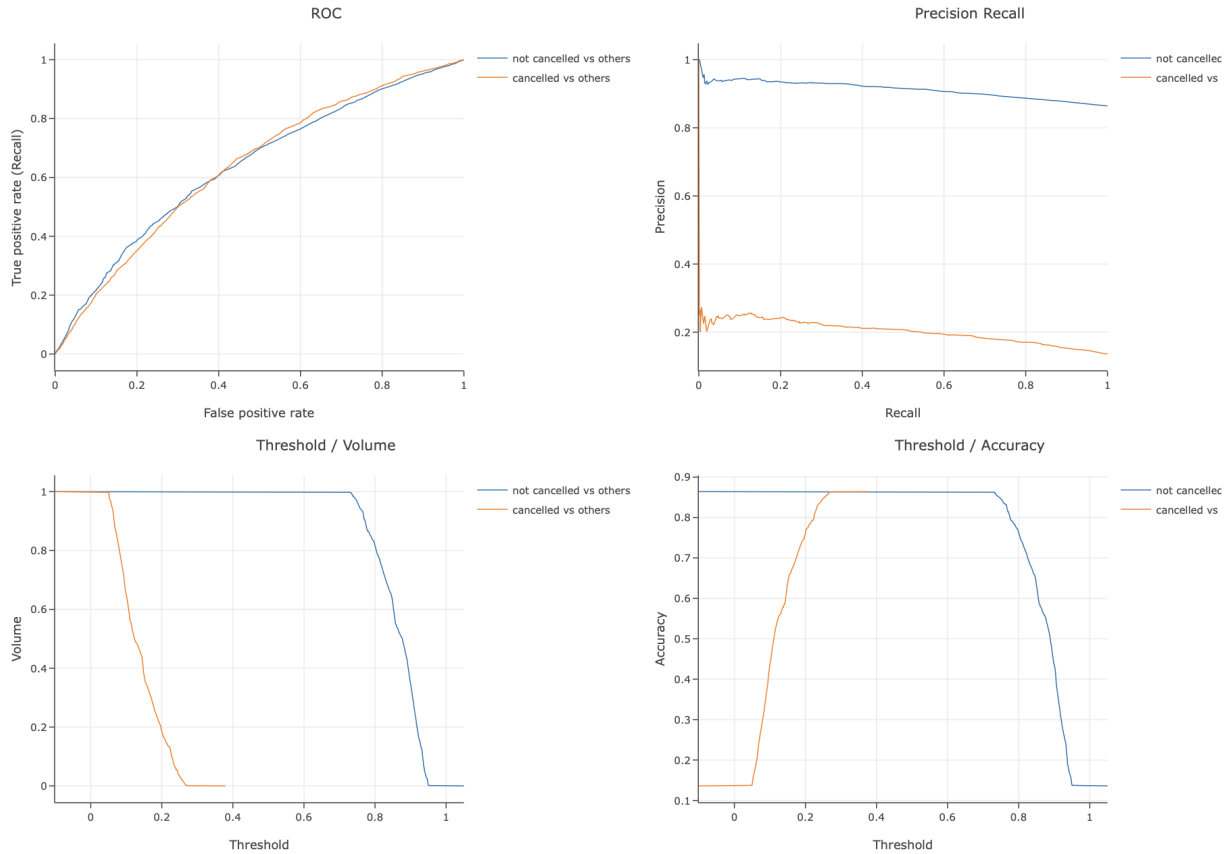
Model Evaluation



Model Evaluation

How to read a model evaluation report

Evaluation report

```
Number of predictions (without weights): 10000
Number of predictions (with weights): 10000
Task: CLASSIFICATION
Label: cancelled

Accuracy: 0.8641  CI95[W][0.858335 0.869708]
LogLoss: : 0.383495
ErrorRate: : 0.1359

Default Accuracy: : 0.8641
Default LogLoss: : 0.39745
Default ErrorRate: : 0.1359

Confusion Table:
truth\prediction
               not cancelled  cancelled
not cancelled           8641          0
    cancelled           1359          0
Total: 10000

  "not cancelled" vs. the others
    auc: 0.639607  CI95[H][0.625026 0.654189]
    p/r-auc: 0.91227  CI95[L][0.906119 0.918054]
    ap: 0.91238

  "cancelled" vs. the others
    auc: 0.639607  CI95[H][0.622834 0.656381]
    p/r-auc: 0.202041  CI95[L][0.181532 0.224232]
    ap: 0.202416
```

Model Understanding

Summary | Quality | Dataset | Variable importance | Predictions | Plot model

How to read a model evaluation report

```
Task: CLASSIFICATION
Label: cancelled
Loss (BINOMIAL_LOG_LIKELIHOOD): 0.808441

Accuracy: 0.851314  CI95[W][0 1]
ErrorRate: : 0.148686


Confusion Table:
truth\prediction
               not cancelled  cancelled
not cancelled           8554          0
    cancelled           1494          0
Total: 10048
```

Model Understanding

Summary | Quality | Dataset | Variable importance | Predictions | Plot model

How to read variable importances

```
Variable Importance: INV_MEAN_MIN_DEPTH:
    1. "monthly_visits"  1.000000 ################
    2.            "age"  0.246628 #
    3.         "gender"  0.178439

Variable Importance: MEAN_DECREASE_IN_ACCURACY:
    1.            "age"  0.000000
    2.         "gender"  0.000000
    3. "monthly_visits"  0.000000

Variable Importance: MEAN_DECREASE_IN_AP_cancelled_VS_OTHERS:
    1. "monthly_visits"  0.076849 ################
    2.            "age" -0.000321
    3.         "gender" -0.001256

Variable Importance: MEAN_DECREASE_IN_AUC_cancelled_VS_OTHERS:
    1. "monthly_visits"  0.143418 ################
    2.         "gender" -0.000068
    3.            "age" -0.000628

Variable Importance: MEAN_DECREASE_IN_PRAUC_cancelled_VS_OTHERS:
    1. "monthly_visits"  0.077253 ################
    2.            "age" -0.000301
    3.         "gender" -0.001190

Variable Importance: NUM_AS_ROOT:
    1. "monthly_visits" 24.000000

Variable Importance: NUM_NODES:
    1.            "age" 438.000000 ################
    2. "monthly_visits" 229.000000 ######
    3.         "gender" 77.000000

Variable Importance: SUM_SCORE:
    1. "monthly_visits" 1938.676051 ################
    2.            "age" 111.293509
    3.         "gender" 13.391405
```